Ledgerline interest calculation: source code and I/O contract

No wiki: the COBOL source of the Ledgerline batch step. CONTRACT.md is the input/output contract for the build.

Pages included:

- `source-code.md`
- `CONTRACT.md`

`source-code.md`

Source code

COBOL source of the Ledgerline ACCRUE step (program LXB002C) with its copybooks and the two runtime shims used to run it.

LXB002C.cbl

```
      ****************************************************************
      * Program     : LXB002C.CBL
      * Application : Ledgerline
      * Type        : BATCH COBOL Program
      * Function    : This is a interest calculator program.
      ****************************************************************
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      ****************************************************************
       IDENTIFICATION DIVISION.
       PROGRAM-ID.    LXB002C.
       AUTHOR.        LEDGERLINE.
       ENVIRONMENT DIVISION.
       INPUT-OUTPUT SECTION.
       FILE-CONTROL.
           SELECT BALCAT-FILE ASSIGN TO BALCATF
                  ORGANIZATION IS INDEXED
                  ACCESS MODE  IS SEQUENTIAL
                  RECORD KEY   IS FD-TXN-CAT-KEY
                  FILE STATUS  IS BALCATF-STATUS.

           SELECT LINK-FILE ASSIGN TO   LINKFILE
                  ORGANIZATION IS INDEXED
                  ACCESS MODE  IS RANDOM
                  RECORD KEY   IS FD-LINK-CRD-NUM
                  ALTERNATE RECORD KEY IS FD-LINK-LACC-ID
                  FILE STATUS  IS LINKFILE-STATUS.

           SELECT ACCOUNT-FILE ASSIGN TO LEDGFILE
                  ORGANIZATION IS INDEXED
                  ACCESS MODE  IS RANDOM
                  RECORD KEY   IS FD-LACC-ID
                  FILE STATUS  IS LEDGFILE-STATUS.

           SELECT RATEGRP-FILE ASSIGN TO RATEGRP
                  ORGANIZATION IS INDEXED
                  ACCESS MODE  IS RANDOM
                  RECORD KEY   IS FD-RATEGRP-KEY
                  FILE STATUS  IS RATEGRP-STATUS.
```

```
           SELECT TXNFILE-FILE ASSIGN TO TXNFILE
                  ORGANIZATION IS SEQUENTIAL
                  ACCESS MODE  IS SEQUENTIAL
                  FILE STATUS  IS TXNFILEX-STATUS.

       *
        DATA DIVISION.
        FILE SECTION.
        FD  BALCAT-FILE.
        01  FD-TXN-CAT-BAL-RECORD.
            05 FD-TXN-CAT-KEY.
               10 FD-BCAT-LACC-ID             PIC 9(11).
               10 FD-BCAT-TYPE-CD             PIC X(02).
               10 FD-BCAT-CD                  PIC 9(04).
            05 FD-FD-TXN-CAT-DATA               PIC X(33).

        FD  LINK-FILE.
        01  FD-LINKFILE-REC.
            05 FD-LINK-CRD-NUM                  PIC X(16).
            05 FD-LINK-CLNT-NUM                  PIC 9(09).
            05 FD-LINK-LACC-ID                   PIC 9(11).
            05 FD-LINK-FILLER                    PIC X(14).

        FD  RATEGRP-FILE.
        01  FD-RATEGRP-REC.
            05 FD-RATEGRP-KEY.
               10 FD-RG-LACC-GROUP-ID            PIC X(10).
               10 FD-RG-TXN-TYPE-CD             PIC X(02).
               10 FD-RG-TXN-CAT-CD              PIC 9(04).
            05 FD-RATEGRP-DATA                    PIC X(34).

        FD  ACCOUNT-FILE.
        01  FD-LEDGFILE-REC.
            05 FD-LACC-ID                         PIC 9(11).
            05 FD-LACC-DATA                       PIC X(289).

        FD  TXNFILE-FILE.
        01  FD-TXNFILEX-REC.
            05 FD-TRANS-ID                        PIC X(16).
            05 FD-LACC-DATA                       PIC X(334).

        WORKING-STORAGE SECTION.

       *****************************************************************
        COPY LXV026Y.
        01  BALCATF-STATUS.
            05  BALCATF-STAT1      PIC X.
            05  BALCATF-STAT2      PIC X.

        COPY LXV024Y.
        01  LINKFILE-STATUS.
            05  LINKFILE-STAT1      PIC X.
            05  LINKFILE-STAT2      PIC X.

        COPY LXV027Y.
        01  RATEGRP-STATUS.
            05 RATEGRP-STAT1        PIC X.
            05 RATEGRP-STAT2        PIC X.
```

```
       COPY LXV022Y.
       01  LEDGFILE-STATUS.
           05  LEDGFILE-STAT1      PIC X.
           05  LEDGFILE-STAT2      PIC X.

       COPY LXV030Y.
       01  TXNFILEX-STATUS.
           05  TXNFILEX-STAT1      PIC X.
           05  TXNFILEX-STAT2      PIC X.

       01  IO-STATUS.
           05  IO-STAT1            PIC X.
           05  IO-STAT2            PIC X.
       01  TWO-BYTES-BINARY        PIC 9(4) BINARY.
       01  TWO-BYTES-ALPHA         REDEFINES TWO-BYTES-BINARY.
           05  TWO-BYTES-LEFT      PIC X.
           05  TWO-BYTES-RIGHT     PIC X.
       01  IO-STATUS-04.
           05  IO-STATUS-0401      PIC 9   VALUE 0.
           05  IO-STATUS-0403      PIC 999 VALUE 0.

       01  APPL-RESULT             PIC S9(9)   COMP.
           88  APPL-AOK            VALUE 0.
           88  APPL-EOF            VALUE 16.

       01  END-OF-FILE             PIC X(01)    VALUE 'N'.
       01  ABCODE                  PIC S9(9) BINARY.
       01  TIMING                  PIC S9(9) BINARY.
      * T I M E S T A M P   D B 2  X(26)     EEEE-MM-DD-UU.MM.SS.HH0000
       01  COBOL-TS.
           05 COB-YYYY                 PIC X(04).
           05 COB-MM                   PIC X(02).
           05 COB-DD                   PIC X(02).
           05 COB-HH                   PIC X(02).
           05 COB-MIN                  PIC X(02).
           05 COB-SS                   PIC X(02).
           05 COB-MIL                  PIC X(02).
           05 COB-REST                 PIC X(05).
       01  DB2-FORMAT-TS               PIC X(26).
       01  FILLER REDEFINES DB2-FORMAT-TS.
           06 DB2-YYYY                 PIC X(004).                     E
           06 DB2-STREEP-1             PIC X.                          -
           06 DB2-MM                   PIC X(002).                     M
           06 DB2-STREEP-2             PIC X.                          -
           06 DB2-DD                   PIC X(002).                     D
           06 DB2-STREEP-3             PIC X.                          -
           06 DB2-HH                   PIC X(002).                     U
           06 DB2-DOT-1                PIC X.
           06 DB2-MIN                  PIC X(002).
           06 DB2-DOT-2                PIC X.
           06 DB2-SS                   PIC X(002).
           06 DB2-DOT-3                PIC X.
           06 DB2-MIL                  PIC 9(002).
           06 DB2-REST                 PIC X(04).
       01 WS-MISC-VARS.
           05 WS-LAST-LACC-NUM         PIC X(11) VALUE SPACES.
           05 WS-MONTHLY-INT           PIC S9(09)V99.
           05 WS-TOTAL-INT             PIC S9(09)V99.
```

```
           05 WS-FIRST-TIME             PIC X(01) VALUE 'Y'.
       01 WS-COUNTERS.
           05 WS-RECORD-COUNT           PIC 9(09) VALUE 0.
           05 WS-TRANID-SUFFIX          PIC 9(06) VALUE 0.

       LINKAGE SECTION.
       01  EXTERNAL-PARMS.
           05  PARM-LENGTH         PIC S9(04) COMP.
           05  PARM-DATE           PIC X(10).
      ****************************************************************
       PROCEDURE DIVISION USING EXTERNAL-PARMS.
           DISPLAY 'START OF EXECUTION OF PROGRAM LXB002C'.
           PERFORM 0000-BALCATF-OPEN.
           PERFORM 0100-LINKFILE-OPEN.
           PERFORM 0200-RATEGRP-OPEN.
           PERFORM 0300-LEDGFILE-OPEN.
           PERFORM 0400-TXNFILEX-OPEN.

           PERFORM UNTIL END-OF-FILE = 'Y'
               IF  END-OF-FILE = 'N'
                   PERFORM 1000-BALCATF-GET-NEXT
                   IF  END-OF-FILE = 'N'
                     ADD 1 TO WS-RECORD-COUNT
                     DISPLAY TXN-CAT-BAL-RECORD
                     IF BCAT-LACC-ID NOT= WS-LAST-LACC-NUM
                       IF WS-FIRST-TIME NOT = 'Y'
                          PERFORM 1050-UPDATE-ACCOUNT
                       ELSE
                          MOVE 'N' TO WS-FIRST-TIME
                       END-IF
                       MOVE 0 TO WS-TOTAL-INT
                       MOVE BCAT-LACC-ID TO WS-LAST-LACC-NUM
                       MOVE BCAT-LACC-ID TO FD-LACC-ID
                       PERFORM 1100-GET-LACC-DATA
                       MOVE BCAT-LACC-ID TO FD-LINK-LACC-ID
                       PERFORM 1110-GET-LINK-DATA
                     END-IF
      *              DISPLAY 'LACC-GROUP-ID: ' LACC-GROUP-ID
      *              DISPLAY 'BCAT-CD: ' BCAT-CD
      *              DISPLAY 'BCAT-TYPE-CD: ' BCAT-TYPE-CD
                     MOVE LACC-GROUP-ID TO FD-RG-LACC-GROUP-ID
                     MOVE BCAT-CD TO FD-RG-TXN-CAT-CD
                     MOVE BCAT-TYPE-CD TO FD-RG-TXN-TYPE-CD
                     PERFORM 1200-GET-INTEREST-RATE
                     IF RG-INT-RATE NOT = 0
                       PERFORM 1300-COMPUTE-INTEREST
                       PERFORM 1400-COMPUTE-FEES
                     END-IF
                   END-IF
               ELSE
                    PERFORM 1050-UPDATE-ACCOUNT
               END-IF
           END-PERFORM.

           PERFORM 9000-BALCATF-CLOSE.
           PERFORM 9100-LINKFILE-CLOSE.
           PERFORM 9200-RATEGRP-CLOSE.
           PERFORM 9300-LEDGFILE-CLOSE.
           PERFORM 9400-TXNFILEX-CLOSE.
```

```
     DISPLAY 'END OF EXECUTION OF PROGRAM LXB002C'.

     GOBACK.
*---------------------------------------------------------------*
 0000-BALCATF-OPEN.
     MOVE 8 TO APPL-RESULT.
     OPEN INPUT BALCAT-FILE
     IF  BALCATF-STATUS = '00'
         MOVE 0 TO APPL-RESULT
     ELSE
         MOVE 12 TO APPL-RESULT
     END-IF
     IF  APPL-AOK
         CONTINUE
     ELSE
         DISPLAY 'ERROR OPENING TRANSACTION CATEGORY BALANCE'
         MOVE BALCATF-STATUS TO IO-STATUS
         PERFORM 9910-DISPLAY-IO-STATUS
         PERFORM 9999-ABEND-PROGRAM
     END-IF
     EXIT.
*---------------------------------------------------------------*
 0100-LINKFILE-OPEN.
     MOVE 8 TO APPL-RESULT.
     OPEN INPUT LINK-FILE
     IF  LINKFILE-STATUS = '00'
         MOVE 0 TO APPL-RESULT
     ELSE
         MOVE 12 TO APPL-RESULT
     END-IF
     IF  APPL-AOK
         CONTINUE
     ELSE
         DISPLAY 'ERROR OPENING CROSS REF FILE'   LINKFILE-STATUS
         MOVE LINKFILE-STATUS TO IO-STATUS
         PERFORM 9910-DISPLAY-IO-STATUS
         PERFORM 9999-ABEND-PROGRAM
     END-IF
     EXIT.
*---------------------------------------------------------------*
 0200-RATEGRP-OPEN.
     MOVE 8 TO APPL-RESULT.
     OPEN INPUT RATEGRP-FILE
     IF  RATEGRP-STATUS = '00'
         MOVE 0 TO APPL-RESULT
     ELSE
         MOVE 12 TO APPL-RESULT
     END-IF
     IF  APPL-AOK
         CONTINUE
     ELSE
         DISPLAY 'ERROR OPENING DALY REJECTS FILE'
         MOVE RATEGRP-STATUS TO IO-STATUS
         PERFORM 9910-DISPLAY-IO-STATUS
         PERFORM 9999-ABEND-PROGRAM
     END-IF
     EXIT.
```

```
       *---------------------------------------------------------------*
        0300-LEDGFILE-OPEN.
            MOVE 8 TO APPL-RESULT.
            OPEN I-O ACCOUNT-FILE
            IF  LEDGFILE-STATUS = '00'
                MOVE 0 TO APPL-RESULT
            ELSE
                MOVE 12 TO APPL-RESULT
            END-IF
            IF  APPL-AOK
                CONTINUE
            ELSE
                DISPLAY 'ERROR OPENING ACCOUNT MASTER FILE'
                MOVE LEDGFILE-STATUS TO IO-STATUS
                PERFORM 9910-DISPLAY-IO-STATUS
                PERFORM 9999-ABEND-PROGRAM
            END-IF
            EXIT.
       *---------------------------------------------------------------*
        0400-TXNFILEX-OPEN.
            MOVE 8 TO APPL-RESULT.
            OPEN OUTPUT TXNFILE-FILE
            IF  TXNFILEX-STATUS = '00'
                MOVE 0 TO APPL-RESULT
            ELSE
                MOVE 12 TO APPL-RESULT
            END-IF
            IF  APPL-AOK
                CONTINUE
            ELSE
                DISPLAY 'ERROR OPENING TRANSACTION FILE'
                MOVE TXNFILEX-STATUS TO IO-STATUS
                PERFORM 9910-DISPLAY-IO-STATUS
                PERFORM 9999-ABEND-PROGRAM
            END-IF
            EXIT.
       *---------------------------------------------------------------*
        1000-BALCATF-GET-NEXT.
            READ BALCAT-FILE INTO TXN-CAT-BAL-RECORD.
            IF  BALCATF-STATUS  = '00'
                MOVE 0 TO APPL-RESULT
            ELSE
                IF  BALCATF-STATUS  = '10'
                    MOVE 16 TO APPL-RESULT
                ELSE
                    MOVE 12 TO APPL-RESULT
                END-IF
            END-IF
            IF  APPL-AOK
                CONTINUE
            ELSE
                IF  APPL-EOF
                    MOVE 'Y' TO END-OF-FILE
                ELSE
                    DISPLAY 'ERROR READING TRANSACTION CATEGORY FILE'
                    MOVE BALCATF-STATUS TO IO-STATUS
                    PERFORM 9910-DISPLAY-IO-STATUS
                    PERFORM 9999-ABEND-PROGRAM
                END-IF
```

```
           END-IF
           EXIT.
      *---------------------------------------------------------------*
       1050-UPDATE-ACCOUNT.
      * Update the balances in account record to reflect posted trans.
           ADD WS-TOTAL-INT  TO LACC-CURR-BAL
           MOVE 0 TO LACC-CURR-CYC-CREDIT
           MOVE 0 TO LACC-CURR-CYC-DEBIT

           REWRITE FD-LEDGFILE-REC FROM  ACCOUNT-RECORD
           IF  LEDGFILE-STATUS  = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF
           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR RE-WRITING ACCOUNT FILE'
               MOVE LEDGFILE-STATUS  TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           EXIT.
      *---------------------------------------------------------------*
       1100-GET-LACC-DATA.
           READ ACCOUNT-FILE INTO ACCOUNT-RECORD
               INVALID KEY
                  DISPLAY 'ACCOUNT NOT FOUND: ' FD-LACC-ID
           END-READ

           IF  LEDGFILE-STATUS  = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF
           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR READING ACCOUNT FILE'
               MOVE LEDGFILE-STATUS  TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           EXIT.
      *---------------------------------------------------------------*
       1110-GET-LINK-DATA.
           READ LINK-FILE INTO CRD-LINK-RECORD
            KEY IS FD-LINK-LACC-ID
               INVALID KEY
                  DISPLAY 'ACCOUNT NOT FOUND: ' FD-LINK-LACC-ID
           END-READ

           IF  LINKFILE-STATUS   = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF
           IF  APPL-AOK
```

```
           CONTINUE
       ELSE
           DISPLAY 'ERROR READING XREF FILE'
           MOVE LINKFILE-STATUS  TO IO-STATUS
           PERFORM 9910-DISPLAY-IO-STATUS
           PERFORM 9999-ABEND-PROGRAM
       END-IF
       EXIT.
      *---------------------------------------------------------*
       1200-GET-INTEREST-RATE.
           READ RATEGRP-FILE INTO RG-GROUP-RECORD
                INVALID KEY
                   DISPLAY 'DISCLOSURE GROUP RECORD MISSING'
                   DISPLAY 'TRY WITH STANDARD GROUP CODE'
           END-READ.

           IF  RATEGRP-STATUS  = '00'  OR '23'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF

           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR READING DISCLOSURE GROUP FILE'
               MOVE RATEGRP-STATUS  TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           IF  RATEGRP-STATUS  = '23'
               MOVE 'STANDARD' TO FD-RG-LACC-GROUP-ID
               PERFORM 1200-A-GET-STANDARD-INT-RATE
           END-IF
           EXIT.

      *---------------------------------------------------------*
       1200-A-GET-STANDARD-INT-RATE.
           READ RATEGRP-FILE INTO RG-GROUP-RECORD

           IF  RATEGRP-STATUS  = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF

           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR READING STANDARD DISCLOSURE GROUP'
               MOVE RATEGRP-STATUS  TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           EXIT.
      *---------------------------------------------------------*
       1300-COMPUTE-INTEREST.

           COMPUTE WS-MONTHLY-INT
```

```
       = ( TXN-CAT-BAL * RG-INT-RATE) / 1200

     ADD WS-MONTHLY-INT  TO WS-TOTAL-INT
     PERFORM 1300-B-WRITE-TX.

     EXIT.

*---------------------------------------------------------------*
 1300-B-WRITE-TX.
     ADD 1 TO WS-TRANID-SUFFIX

     STRING PARM-DATE,
            WS-TRANID-SUFFIX
       DELIMITED BY SIZE
       INTO TXN-ID
     END-STRING.

     MOVE '07'                 TO TXN-TYPE-CD
     MOVE '12'                 TO TXN-CAT-CD
     MOVE 'Batch'              TO TXN-SOURCE
     STRING 'Accrued int ' ,
            LACC-ID
            DELIMITED BY SIZE
      INTO TXN-DESC
     END-STRING
     MOVE WS-MONTHLY-INT       TO TXN-AMT
     MOVE 0                    TO TXN-MERCHANT-ID
     MOVE SPACES               TO TXN-MERCHANT-NAME
     MOVE SPACES               TO TXN-MERCHANT-CITY
     MOVE SPACES               TO TXN-MERCHANT-ZIP
     MOVE LINK-CRD-NUM        TO TXN-CRD-NUM
     PERFORM Z-GET-DB2-FORMAT-TIMESTAMP
     MOVE DB2-FORMAT-TS        TO TXN-ORIG-TS
     MOVE DB2-FORMAT-TS        TO TXN-PROC-TS

     WRITE FD-TXNFILEX-REC FROM TXN-RECORD
     IF  TXNFILEX-STATUS   = '00'
         MOVE 0 TO APPL-RESULT
     ELSE
         MOVE 12 TO APPL-RESULT
     END-IF

     IF  APPL-AOK
         CONTINUE
     ELSE
         DISPLAY 'ERROR WRITING TRANSACTION RECORD'
         MOVE TXNFILEX-STATUS   TO IO-STATUS
         PERFORM 9910-DISPLAY-IO-STATUS
         PERFORM 9999-ABEND-PROGRAM
     END-IF
     EXIT.

*---------------------------------------------------------------*
 1400-COMPUTE-FEES.
* To be implemented
     EXIT.
*---------------------------------------------------------------*
 9000-BALCATF-CLOSE.
     MOVE 8 TO  APPL-RESULT.
```

```
           CLOSE BALCAT-FILE
           IF  BALCATF-STATUS = '00'
               MOVE 0 TO  APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF
           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR CLOSING TRANSACTION BALANCE FILE'
               MOVE BALCATF-STATUS  TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           EXIT.

      *---------------------------------------------------------------*
       9100-LINKFILE-CLOSE.
           MOVE 8 TO APPL-RESULT.
           CLOSE LINK-FILE
           IF  LINKFILE-STATUS = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF
           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR CLOSING CROSS REF FILE'
               MOVE LINKFILE-STATUS TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           EXIT.
      *---------------------------------------------------------------*
       9200-RATEGRP-CLOSE.
           MOVE 8 TO APPL-RESULT.
           CLOSE RATEGRP-FILE
           IF  RATEGRP-STATUS = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
               MOVE 12 TO APPL-RESULT
           END-IF
           IF  APPL-AOK
               CONTINUE
           ELSE
               DISPLAY 'ERROR CLOSING DISCLOSURE GROUP FILE'
               MOVE RATEGRP-STATUS TO IO-STATUS
               PERFORM 9910-DISPLAY-IO-STATUS
               PERFORM 9999-ABEND-PROGRAM
           END-IF
           EXIT.
      *---------------------------------------------------------------*
       9300-LEDGFILE-CLOSE.
           MOVE 8 TO APPL-RESULT.
           CLOSE ACCOUNT-FILE
           IF  LEDGFILE-STATUS  = '00'
               MOVE 0 TO APPL-RESULT
           ELSE
```

```
            MOVE 12 TO APPL-RESULT
        END-IF
        IF  APPL-AOK
            CONTINUE
        ELSE
            DISPLAY 'ERROR CLOSING ACCOUNT FILE'
            MOVE LEDGFILE-STATUS  TO IO-STATUS
            PERFORM 9910-DISPLAY-IO-STATUS
            PERFORM 9999-ABEND-PROGRAM
        END-IF
        EXIT.

     9400-TXNFILEX-CLOSE.
        MOVE 8 TO APPL-RESULT.
        CLOSE TXNFILE-FILE
        IF  TXNFILEX-STATUS  = '00'
            MOVE 0 TO APPL-RESULT
        ELSE
            MOVE 12 TO APPL-RESULT
        END-IF
        IF  APPL-AOK
            CONTINUE
        ELSE
            DISPLAY 'ERROR CLOSING TRANSACTION FILE'
            MOVE TXNFILEX-STATUS  TO IO-STATUS
            PERFORM 9910-DISPLAY-IO-STATUS
            PERFORM 9999-ABEND-PROGRAM
        END-IF
        EXIT.

     Z-GET-DB2-FORMAT-TIMESTAMP.
        MOVE FUNCTION CURRENT-DATE TO COBOL-TS
        MOVE COB-YYYY TO DB2-YYYY
        MOVE COB-MM   TO DB2-MM
        MOVE COB-DD   TO DB2-DD
        MOVE COB-HH   TO DB2-HH
        MOVE COB-MIN  TO DB2-MIN
        MOVE COB-SS   TO DB2-SS
        MOVE COB-MIL  TO DB2-MIL
        MOVE '0000'   TO DB2-REST
        MOVE '-' TO DB2-STREEP-1 DB2-STREEP-2 DB2-STREEP-3
        MOVE '.' TO DB2-DOT-1 DB2-DOT-2 DB2-DOT-3
    *    DISPLAY 'DB2-TIMESTAMP = ' DB2-FORMAT-TS
        EXIT.

     9999-ABEND-PROGRAM.
        DISPLAY 'ABENDING PROGRAM'
        MOVE 0 TO TIMING
        MOVE 999 TO ABCODE
        CALL 'LXZ009D' USING ABCODE, TIMING.

    ***************************************************************
     9910-DISPLAY-IO-STATUS.
        IF  IO-STATUS NOT NUMERIC
        OR  IO-STAT1 = '9'
            MOVE IO-STAT1 TO IO-STATUS-04(1:1)
            MOVE 0        TO TWO-BYTES-BINARY
            MOVE IO-STAT2 TO TWO-BYTES-RIGHT
            MOVE TWO-BYTES-BINARY TO IO-STATUS-0403
```

```
           DISPLAY 'FILE STATUS IS: NNNN' IO-STATUS-04
       ELSE
           MOVE '0000' TO IO-STATUS-04
           MOVE IO-STATUS TO IO-STATUS-04(3:2)
           DISPLAY 'FILE STATUS IS: NNNN' IO-STATUS-04
       END-IF
       EXIT.

      *
      * Ver: Ledgerline_v2.0-25-gdb72e6b-235 Date: 2025-04-29 11:01:28 CDT
      *
```

LXV022Y.cpy

```
      ******************************************************************
      *    Data-structure for  account entity (RECLN 300)
      ******************************************************************
       01  ACCOUNT-RECORD.
           05  LACC-ID                             PIC 9(11).
           05  LACC-ACTIVE-STATUS                  PIC X(01).
           05  LACC-CURR-BAL                       PIC S9(10)V99.
           05  LACC-CREDIT-LIMIT                   PIC S9(10)V99.
           05  LACC-CASH-CREDIT-LIMIT              PIC S9(10)V99.
           05  LACC-OPEN-DATE                      PIC X(10).
           05  LACC-EXPIRY-DATE                  PIC X(10).
           05  LACC-REISSUE-DATE                   PIC X(10).
           05  LACC-CURR-CYC-CREDIT                PIC S9(10)V99.
           05  LACC-CURR-CYC-DEBIT                 PIC S9(10)V99.
           05  LACC-ADDR-ZIP                       PIC X(10).
           05  LACC-GROUP-ID                       PIC X(10).
           05  FILLER                              PIC X(178).
      *
      * Ver: Ledgerline_v1.0-15-g27d6c6f-68 Date: 2022-07-19 23:15:59 CDT
      *
```

LXV024Y.cpy

```
      ******************************************************************
      *    Data-structure for card xref (RECLN 50)
      ******************************************************************
       01 CRD-LINK-RECORD.
           05  LINK-CRD-NUM                      PIC X(16).
           05  LINK-CLNT-ID                       PIC 9(09).
           05  LINK-LACC-ID                       PIC 9(11).
           05  FILLER                             PIC X(14).
      *
      * Ver: Ledgerline_v1.0-15-g27d6c6f-68 Date: 2022-07-19 23:16:00 CDT
      *
```

LXV026Y.cpy

```
      ****************************************************************
      *    Data-structure for transaction category balance (RECLN = 50)
      ****************************************************************
       01  TXN-CAT-BAL-RECORD.
           05  TXN-CAT-KEY.
              10 BCAT-LACC-ID                      PIC 9(11).
              10 BCAT-TYPE-CD                      PIC X(02).
              10 BCAT-CD                           PIC 9(04).
           05  TXN-CAT-BAL                          PIC S9(09)V99.
           05  FILLER                                PIC X(22).
      *
      * Ver: Ledgerline_v1.0-15-g27d6c6f-68 Date: 2022-07-19 23:16:00 CDT
      *
```

LXV027Y.cpy

```
      ****************************************************************
      *    Data-structure for disclosure group (RECLN = 50)
      ****************************************************************
       01  RG-GROUP-RECORD.
           05  RG-GROUP-KEY.
              10 RG-LACC-GROUP-ID                   PIC X(10).
              10 RG-TXN-TYPE-CD                    PIC X(02).
              10 RG-TXN-CAT-CD                     PIC 9(04).
           05  RG-INT-RATE                          PIC S9(04)V99.
           05  FILLER                                PIC X(28).
      *
      * Ver: Ledgerline_v1.0-15-g27d6c6f-68 Date: 2022-07-19 23:16:00 CDT
      *
```

LXV030Y.cpy

```
      ****************************************************************
      *    Data-structure for TRANsaction record (RECLN = 350)
      ****************************************************************
       01  TXN-RECORD.
           05  TXN-ID                               PIC X(16).
           05  TXN-TYPE-CD                          PIC X(02).
           05  TXN-CAT-CD                           PIC 9(04).
           05  TXN-SOURCE                           PIC X(10).
           05  TXN-DESC                             PIC X(100).
           05  TXN-AMT                              PIC S9(09)V99.
           05  TXN-MERCHANT-ID                      PIC 9(09).
           05  TXN-MERCHANT-NAME                    PIC X(50).
           05  TXN-MERCHANT-CITY                    PIC X(50).
           05  TXN-MERCHANT-ZIP                     PIC X(10).
           05  TXN-CRD-NUM                         PIC X(16).
           05  TXN-ORIG-TS                          PIC X(26).
           05  TXN-PROC-TS                          PIC X(26).
           05  FILLER                                PIC X(20).
      *
      * Ver: Ledgerline_v1.0-15-g27d6c6f-68 Date: 2022-07-19 23:16:01 CDT
      *
```

LXZ009D.cbl

```
       IDENTIFICATION DIVISION.
      * Stub for the z/OS Language Environment abend service.
       PROGRAM-ID. LXZ009D.
       DATA DIVISION.
       LINKAGE SECTION.
       01 ABCODE PIC S9(9) BINARY.
       01 TIMING PIC S9(9) BINARY.
       PROCEDURE DIVISION USING ABCODE TIMING.
           DISPLAY 'LXZ009D ABEND CALLED'
           STOP RUN RETURNING 16.
```

LXZ036M.cbl

```
       IDENTIFICATION DIVISION.
      * Driver: emulates JCL EXEC PGM=<prog>,PARM='...' for a program
      * whose PROCEDURE DIVISION takes the z/OS PARM block.
       PROGRAM-ID. LXZ036M.
       DATA DIVISION.
       WORKING-STORAGE SECTION.
       01 WS-PROG   PIC X(8).
       01 WS-ARG    PIC X(100).
       01 PARM-BLOCK.
          05 PARM-LENGTH PIC S9(4) COMP.
          05 PARM-TEXT   PIC X(100).
       PROCEDURE DIVISION.
           ACCEPT WS-PROG FROM ARGUMENT-VALUE
           ACCEPT WS-ARG  FROM ARGUMENT-VALUE
           MOVE WS-ARG TO PARM-TEXT
           MOVE FUNCTION LENGTH(FUNCTION TRIM(WS-ARG)) TO PARM-LENGTH
           CALL WS-PROG USING PARM-BLOCK
           STOP RUN RETURNING RETURN-CODE.
```

Candidate I/O contract

What a migrated implementation must produce so the scorer can compare it with the oracle. The contract fixes **formats only**. It says nothing about behaviour, so it may be handed to a builder without leaking the answers.

Inputs (given per run)

The same files the COBOL chain reads, one record per line, fixed width, layouts as in the copybooks:

File	Copybook	Record
`daytxn.dat`	LXV031Y	350
`ledgacc.dat`	LXV022Y	300
`cardlink.dat`	LXV024Y	50
`balcat.dat`	LXV026Y	50
`rategrp.dat`	LXV027Y	50
`clntrec.dat`, `crdrec.dat`, `txntype.dat`, `txncatg.dat`	LXU021C, LXV023Y, LXV028Y, LXV029Y	500, 150, 60, 60

Run parameters: business date `2022-07-18`, ACCRUE parameter `2022071800`.

Step isolation. A candidate for ACCRUE receives the oracle's POSTDAY outputs as its account and category-balance inputs, so an ACCRUE result is never penalised for a POSTDAY error.

Outputs

One JSON file per output: a list of objects, in any order. Field names and types are fixed. Strings may omit trailing spaces. Amounts are JSON strings or numbers holding at most two decimals (`"14.56"`, `-500`).

File	Step	Fields
`postday_txnfile.json`	POSTDAY	transaction fields (below) for every posted transaction
`postday_dayrej.json`	POSTDAY	transaction fields + `reason` (int) + `reason_desc` for every rejected transaction
`postday_ledgacc.json`	POSTDAY	account fields for every account, after posting
`postday_balcat.json`	POSTDAY	`acct_id, type_cd, cat_cd, bal` for every category balance, after posting
`accrue_systxn.json`	ACCRUE	transaction fields for every interest transaction written
`accrue_ledgacc.json`	ACCRUE	account fields for every account, after interest

- Transaction fields: `tran_id, type_cd, cat_cd (int), source, desc, amt, merchant_id (int), merchant_name, merchant_city, merchant_zip, card_num, orig_ts, proc_ts`
- Account fields: `acct_id (int), active_status, curr_bal, credit_limit, cash_credit_limit, open_date, expiration_date, reissue_date, curr_cyc_credit, curr_cyc_debit, addr_zip, group_id`

A run may instead leave the oracle's fixed-width dumps (`<name>.txt`). The scorer reads either form.

Optional `flags.json`: the uncertainties the pipeline declared **before** scoring, as `[{"rule": "R-INT-TRUNC", "source": "<artefact path and line>"}]` or `[{"output": "...", "key": "...", "source": "..."}]`. See PROTOCOL.md §6 for what counts as a flag.